

Group 5 Proposal

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1. Motivation of the project

Spatial transcriptomics opens a new window into tissue biology by revealing gene expression patterns in their native spatial context. A foundational problem is the identification of Spatially Variable Genes (SVGs). SpatialDE (Svensson et al., 2018) introduced the first statistical framework for this task, utilizing a Gaussian process model to decompose each gene's variability into spatial versus non-spatial components.

However, SpatialDE was developed during the infancy of spatial technologies. Its reliance on Gaussian assumptions limits its theoretical efficacy on modern platforms, which generate sparse, zero-inflated count data with complex cell-type mixtures. Furthermore, the original benchmark studies for both SpatialDE and newer Poisson-based methods like SPARK primarily evaluated their models using early, low-resolution datasets or animal models. Consequently, their accuracy, scalability, and reliability on contemporary high-resolution platforms and complex human clinical tissues remain largely unverified.

To bridge this critical gap, we propose a systematic benchmark of SpatialDE against SPARK, explicitly focusing on diverse, high-resolution human spatial datasets. By quantifying statistical power, False Discovery Rate (FDR), computational scalability, and the biological relevance of detected gene sets, this project will update the legacy of these foundational tools. Ultimately, this analysis will establish robust guidelines for applying SVG algorithms to modern, high-density clinical data and inform future methodological developments.

2. Description of Methodology

To quantify the differences between newer SVG identification methods, we will be comparing SpatialDE and SPARK. We will examine diverse high resolution human spatial transcriptomics datasets, and rank genes via SPARK and SpatialDE. We will compare the results of both methods via their statistical correlation, sensitivity to different types of datasets, and utility in determining gene expression.

3. Dataset and Data Processing

For this project, the following datasets will be used to provide a comprehensive evaluation:

- **Mouse Olfactory Bulb (Low Resolution):** We will use the Ståhl et al. (2016) dataset, which consists of approximately 260 spots and 16,000 genes. This was the primary validation case in the original SpatialDE research, which acts as a strong baseline for reproducing the authors' initial findings.
- **Mouse Hippocampus (High Resolution):** To simulate current challenges, we will use Slide-seq V2 data, derived from Shah et al.'s seqFISH dataset. This will act as the primary stress test for SpatialDE's Gaussian assumptions, compared to Poisson-based alternatives like SPARK.
- **Human Breast Cancer (Heterogeneous Tissue):** To fully test both models' ability to detect spatial variability, tissue slices from human breast cancer biopsies from Ståhl et al. will be used. Using data from a clinically relevant, non uniform environment with technical noise, will provide a strong conclusion on model reactions.

To satisfy the statistical requirements of both Gaussian Process (SpatialDE) and Generalized Linear Spatial Model (SPARK) frameworks, a designated preprocessing workflow will be designed. Spots with fewer than 10 total Unique Molecular Identifier (UMI) counts will be removed from the data. Genes expressed in less than 10% of total spatial locations will also be excluded. This ensures that the covariance matrix will have sufficient non-zero entries to accurately calculate metrics.

SpatialDE and SPARK assume different underlying distributions, so the data will be transformed differently for each method. For SpatialDE, data will be normalized and then log-transformed to stabilize variance to align with the Gaussian noise model. SPARK models a Poisson distribution, so data will maintain raw UMI counts. To ensure consistency, all spatial coordinates will be scaled to a range of [0,1]. This standardization will allow us to test a uniform range of spatial length-scales without being biased by the units of the original imaging platform.

4. Implementation and Evaluation

4.1 Implementation

For our implementation we will run SpatialDE using its primary Python package, while SPARK will be evaluated within an R environment. To ensure that our performance metrics are directly comparable, we will standardize both implementations to run on the identical computational architecture. As detailed in Section 3, data will be preprocessed and normalized according to each model's specific statistical assumptions.

Finally to guarantee absolute replicability of this project, we will execute all data processing and model initializations using a globally fixed random seed. During execution, we will automatically record computational runtime and peak memory usage in the background to evaluate practical scalability without altering the model's workflow.

Upon completion of a run, both models return specific dataframes containing the spatial variability statistics needed for our evaluation. Specifically, the `SpatialDE.run()` function outputs a pandas dataframe containing the gene name, the raw p-value, the multiple-testing corrected significance (qval), and the spatial distance scale. Conversely, the SPARK framework outputs a result matrix that includes the combined p-value and adjusted p-value for each gene. To ensure a standardized comparison during our evaluation phase, we will explicitly extract the SpatialDE qval and SPARK's adjusted p-value to serve as our equivalent metrics for filtering significant genes.

4.2 Evaluation

To see how well these models perform under different conditions and assess their practical utility, we will evaluate their outputs across four different categories:

- **Statistical Power:** To evaluate initial detection capabilities, we will define significant spatially variable genes (SVG) using an adjusted p-value threshold of < 0.05 . We will compare the total yield between SpatialDE and SPARK, and measure how much the models agree on their prioritizations of specific genes using pairwise Spearman's correlation.
- **False Discovery Rate (FDR) and Sensitivity to Data Sparsity:** Because modern spatial datasets are highly sparse we will explicitly test how SpatialDE gaussian assumptions compare to SPARK's Poisson model when handling low-count data. By randomly down-sampling the spatial points in our high resolution datasets by 80% and rerunning both models, we will estimate the FDR. We define this as the proportion of significant

SVG's uniquely detected in the subsampled data that were not present in the original full dataset.

- **Biological Relevance:** To determine if the mathematically detected SVG's translate to actual biological insights, we will evaluate their utility using standard spatial domain clusters. We will select a fixed number of top SVG's (e.g., the top 500) identified by both models to perform a baseline k-means clustering. We will then quantify how well these clusters reconstruct the known histological layers using the adjusted rand index (ARI).
- **Computational Scalability:** Finally to assess the practical viability of running these models on local hardware, we will compare their execution runtime and peak memory footprint as the dataset size scales.

5. Role Description

Sapna Senthil, Data Processing: The data will need to be cleaned and normalized.

Low-quality observations will be removed from the dataset. Then, copies of the data will be created before normalization. Each copy of the data will be normalized and transformed to better fit either SpatialDE or SPARK. After these steps, the data will be ready for testing.

Dean Catlin, SpatialDE Programming: The data will need to be tested using both SpatialDE, Dean will run SpatialDE on the dataset for comparison with the SPARK results.

Bilal Ahmed, SPARK Programming: The data will need to be tested using both SpatialDE, Bilal will run SPARK on the dataset for comparison with the SpatialDE results.

Xiaoyao Li, Result Evaluation and Biological Interpretation: After both models generate their results, the outputs will be systematically compared. Xiaoyao will assess the statistical performance and biological relevance of the identified SVGs from both SpatialDE and SPARK to determine their practical efficacy.

References

Svensson, V., Teichmann, S. & Stegle, O. SpatialDE: identification of spatially variable genes. *Nat Methods* 15, 343–346 (2018). <https://doi.org/10.1038/nmeth.4636>

Ståhl et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* Vol 353, Issue 6294, 78-82 (2016). <https://doi.org/10.1126/science.aaf2403>